

PHYS / CS 405

CS 405: Quantum Computing Notes

Comprehensive Lecture Notes & Quantum Circuit Primer

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## LECTURE 1

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## Qubits, Superposition &amp; State Vectors

Welcome to Quantum Information & Computing! In this introductory lecture, we explore the fundamental mathematical transition from classical binary bits to quantum state vectors in Hilbert space.

## 1 Classical Bits vs. Quantum Bits

A classical bit exists in one of two definite states: 0 or 1. A quantum bit (qubit), by contrast, is a two-level quantum system represented as a unit vector in a 2-dimensional complex vector space  $\mathbb{C}^2$ .

### [DEF] Definition 1 (Qubit & Computational Basis)

The computational basis states of a qubit are orthonormal basis vectors:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

A general qubit state  $|\psi\rangle$  is a linear superposition:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad \alpha, \beta \in \mathbb{C}$$

subject to the **normalization condition**:  $|\alpha|^2 + |\beta|^2 = 1$ .

### [THM] Theorem 1 (Born Rule for Measurement)

When a qubit in state  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$  is measured in the computational basis, the outcome is:

- State  $|0\rangle$  with probability  $P(0) = |\alpha|^2$
- State  $|1\rangle$  with probability  $P(1) = |\beta|^2$

Immediately following measurement, the qubit collapses state to the observed state vector.

**Proof:** By projection operator measurement postulates, the probability of outcome  $i \in \{0, 1\}$  is given by  $P(i) = \langle\psi|P_i|\psi\rangle$  where  $P_0 = |0\rangle\langle 0|$  and  $P_1 = |1\rangle\langle 1|$ . Substituting  $|\psi\rangle$  yields  $P(0) = |\alpha|^2$  and  $P(1) = |\beta|^2$ . ■

### [EX] Example 1 (Equal Superposition State)

The Hadamard state  $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$  has  $\alpha = \beta = 1/\sqrt{2}$ . Measuring  $|+\rangle$  yields outcome 0 with probability  $|\frac{1}{\sqrt{2}}|^2 = 0.5$  and outcome 1 with probability 0.5.

**[!] Warning**

Quantum superposition is not simply an unknown classical state! A qubit in state  $|+\rangle$  possesses definitive quantum coherence that can produce quantum interference when transformed by unitary gates.

**Python Simulation: Qubit Superposition with NumPy**

```

1 import numpy as np
2
3 # Define computational basis vectors
4 ket_0 = np.array([[1], [0]])
5 ket_1 = np.array([[0], [1]])
6
7 # Define Hadamard matrix H
8 H = (1 / np.sqrt(2)) * np.array([[1, 1],
9                                   [1, -1]])
10
11 # Prepare state  $|+\rangle = H|0\rangle$ 
12 ket_plus = H @ ket_0
13 print("State  $|+\rangle$ :\n", ket_plus)
14
15 # Probabilities  $P(0)$  and  $P(1)$ 
16 prob_0 = np.abs(ket_plus[0, 0])**2
17 prob_1 = np.abs(ket_plus[1, 0])**2
18 print(f"P(0) = {prob_0:.2f}, P(1) = {prob_1:.2f}")

```

**[KEY] Key Takeaways**

- Qubits live in  $\mathbb{C}^2$  complex Hilbert space normalized to length 1.
- Measurement collapses superposition non-reversibly into basis states.
- Unitary operations preserve norms and quantum coherence.

## LECTURE 2

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## Entanglement &amp; Bell States

Quantum entanglement is a non-local correlation between multiple qubits that cannot be factored into product states of individual qubits.

## 2 Multi-Qubit Systems & Tensor Products

A 2-qubit system lives in the tensor product space  $\mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4$  with computational basis  $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ .

### [DEF] Definition 2 (Product vs. Entangled States)

A two-qubit state  $|\Psi\rangle$  is a **product state** if it can be written as  $|\Psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$ . Otherwise,  $|\Psi\rangle$  is **entangled**.

### [THM] Theorem 2 (Maximally Entangled Bell State)

The EPR pair (Bell state  $|\Phi^+\rangle$ ) defined by:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

is a maximally entangled state. Measuring the first qubit collapses the second qubit instantly to the identical state, regardless of spatial distance.

### [\*] Pro Tip

To test if a 2-qubit state  $|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$  is separable (a product state), check whether its coefficient matrix determinant  $ad - bc = 0$ .

### [EXR] Exercise 1 (Bell State Measurement)

Calculate the probability of measuring state  $|01\rangle$  for the Bell state  $|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$ .

### Solution:

The coefficient of  $|01\rangle$  is  $\frac{1}{\sqrt{2}}$ . Therefore, by the Born Rule:

$$P(01) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} = 50\%$$